



The equation of a straight line

1 Match the keywords to their definitions. Write the correct letter in each box

a	constant
b	coefficient
c	gradient
d	y-intercept
e	origin
f	equation of a straight line

	a numerical multiplier of a specific variable in a term
	a measure of how steep a line is
	the coordinate where a line crosses the y axis
	the coordinate (0, 0)
	a relationship which, when plotted, forms a straight line
	a term that does not vary

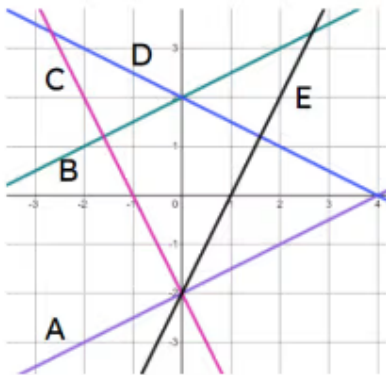
2 Which of these relationships are represented by this bar model? Tick **3** correct answers

y	5
x	

- ☐ $y - x = 5$
☐ $y + x = 5$
☐ $x - y = 5$
☐ $x = y + 5$
☐ $y = x - 5$



- 3 The graph shows 5 lines labelled A, B, C, D and E. Match the line to its gradient and y intercept. Write the correct letter in each box



a	A (purple)
b	B (green)
c	C (pink)
d	D (blue)
e	E (black)

	gradient 2, y-intercept (0, - 2)
	gradient - 2, y-intercept (0, - 2)
	gradient $\frac{1}{2}$, y-intercept (0, - 2)
	gradient $\frac{1}{2}$, y-intercept (0, 2)
	gradient $-\frac{1}{2}$, y-intercept (0, 2)

- 4 Match the equations of lines to their key features. Write the correct letter in each box

a	$y = 5x - 4$
b	$y = 4 - 5x$
c	$y = -4x - 5$
d	$y = 4x + 5$
e	$y = 5x + 4$
f	$y = 5 - 4x$

	gradient 5, y-intercept (0, - 4)
	gradient - 4, y-intercept (0, - 5)
	gradient 4, y-intercept (0, 5)
	gradient - 5, y-intercept (0, 4)
	gradient 5, y-intercept (0, 4)
	gradient - 4, y-intercept (0, 5)



5 What are the key features of the graph with equation $y = 3(5 - 2x)$? Tick **1** correct answer

☐ gradient 15, y-intercept $(0, -6)$

☐ gradient 6, y-intercept $(0, -15)$

☐ gradient -2 , y-intercept $(0, 15)$

☐ gradient -2 , y-intercept $(0, 5)$

☐ gradient -6 , y-intercept $(0, 15)$

6 Which coordinates are on the line $2x - y = 5$? Tick **2** correct answers

☐ $(-2, -9)$

☐ $(-1, 3)$

☐ $(0, 5)$

☐ $(7, 9)$

☐ $(10, 5)$

